Exercise Session – Chapter 4: SDN, NFV, IoT

Exercise 1
Explain the difference between the **Control Plane** and **Data Plane** in SDN.
Then describe what happens when a switch receives a packet **without a matching flow entry**.

Exercise 2

Given the SDN architecture (Application Layer, Control Layer, Data Layer), identify where the following components belong:

- Load balancing algorithm
- OpenFlow switch
- SDN Controller
- Firewall policy (defined by an admin)
- Northbound API call

And justify each answer.

Exercise 3
A switch has the following flow entries:

Match	Action
src=10.1.1.0/24	forward to port 2
dst=192.168.10.5	drop
protocol=TCP	forward to port 1

A packet arrives with:

- src = 10.1.1.3
- dst = 192.168.10.5
- protocol = TCP

Which rule is applied, and why?

Exercise 4
Give 3 advantages and 3 security risks of SDN in enterprise networks.
For each risk, propose one mitigation technique.

Exercise 5
An ISP wants to replace physical firewalls with virtual firewalls (VNFs).
Explain **why NFV is beneficial** in terms of:

- cost
- deployment time
- elasticity
- maintenance

- scalability

Exercise 6
You are asked to design a service chain for a secure video streaming service.
Available VNFs:

- VPN Gateway
- NAT
- Firewall
- Traffic Optimizer
- DPI Engine

Choose **4 VNFs**, put them in an order, and justify your ordering.

Exercise 7
Map the following components to their correct **IoT 5-layer architecture layer** (Perception, Network, Middleware, Application, Business):

- Mobile app showing temperature
- RFID temperature sensor
- MQTT Broker
- IPv6 + 6LoWPAN
- Cloud analytics generating alerts

Exercise 8

A smart city uses IoT devices, NFV-based security services, and SDN control.
Describe how SDN and NFV together ensure:

- efficient routing,
- scalable security,
- dynamic service provisioning for thousands of IoT sensors.

Solutions

Exercise 1

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- The Control Plane decides *how packets should be forwarded*. In SDN, this intelligence is centralized in the SDN Controller.
- The Data Plane forwards packets based on the rules installed by the controller.

When a switch receives a packet with no matching flow entry:

- It sends a Packet-In message to the SDN Controller.
- The controller analyzes the packet and decides what to do.
- It installs a Flow-Mod rule into the switch.
- The switch then forwards future matching packets without contacting the controller again.

Exercise 2

Component	Layer	Explanation
Load balancing algorithm	Application Layer	It defines policies but does not control hardware directly.
OpenFlow switch	Data/Infrastructure Layer	Performs packet forwarding.
SDN Controller	Control Layer	Makes forwarding decisions and manages switches.
Firewall policy	Application Layer	High-level policies defined by admins.
Northbound API call	Between App → Controller	Used by applications to request network services from the controller.

Exercise 3

Packet:

- src=10.1.1.3 (matches Rule 1)
- dst=192.168.10.5 (matches Rule 2)
- protocol=TCP (matches Rule 3)

Which rule applies?
→ Rule 2 (drop)

Why?
OpenFlow uses priority.
A rule with a more specific match gets higher precedence.

- Matching a specific destination IP (dst=192.168.10.5) is more specific than matching a subnet or matching only protocol.

Therefore, the packet is dropped.

Exercise 4

Advantages:

1. Centralized network management → easier automation.
2. Programmability → dynamic changes to flow tables.
3. Faster innovation → new services do not require hardware changes.

Risks and Mitigations:

- Single point of failure (controller outage)
  → Use distributed controllers / clustering.
- Attacks on northbound APIs
  → Use authentication + HTTPS + access control.
- Flow rule spoofing / fake switches
  → Mutual TLS between switches and controller.

Exercise 5

Why NFV is beneficial:

- Cost: Uses commodity servers, no need for expensive appliances.
- Deployment time: VNFs deploy in minutes, not months.
- Elasticity: VNFs scale up/down based on demand.
- Maintenance: Updates and patches done centrally → no physical intervention.
- Scalability: Easy to add multiple VNF instances dynamically.

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Exercise 6

Example correct chain:

VPN Gateway → Firewall → DPI → Traffic Optimizer

Justification:

- VPN first: decrypts traffic so security devices can inspect it.
- Firewall second: blocks unauthorized traffic early.
- DPI third: inspects packet content once decrypted and filtered.
- Traffic Optimizer last: compresses, prioritizes, or shapes traffic after it’s safe.

Other valid chains exist if logical justification is correct.

Exercise 7

Component	Layer
Mobile app showing temperature	Application Layer
RFID temperature sensor	Perception Layer
MQTT Broker	Middleware Layer
IPv6 + 6LoWPAN	Network Layer
Cloud analytics	Business Layer

Exercise 8

Solution summary:

- SDN dynamically routes IoT traffic, prioritizing critical data (e.g., alarms).
- NFV hosts firewalls, DPI, and IDS as VNFs, scaling them as IoT traffic grows.
- Together: traffic from IoT sensors goes through a chain of VNFs, and SDN ensures efficient, programmable routing across the smart city.